

Lorenz Zwick

Research Fellow at the Niels Bohr International Academy
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Curriculum Vitae

Scientific Interests

I am interested in several areas of astrophysics related to **black holes**: The formation and growth of **quasars**, approximation schemes for **analytical gravitational waveforms** and the importance of **environmental effects** for future gravitational wave detectors. I am also interested in the possibility of detecting gravitational waves and dark matter with **Doppler ranging** missions in the Solar system.

Education

Ph.D in theoretical astrophysics. September 2019 - September 2023
At the CTAC, University of Zürich.
Under the supervision of Prof. Lucio Mayer.

Masters Degree in physics. September 2017 - June 2019
At the Eidgenössische Technische Hochschule Zürich.

Bachelor Degree in physics. September 2014 - June 2017
At the Eidgenössische Technische Hochschule Zürich.

Languages and IT Skills

Fluent in Italian, English and German.
Intermediate level (B1 - B2) in French.
Beginner level (A2) in Danish.

Fluent in Python and Mathematica.
Experience in coding **direct integrators for N-body systems** with post-Newtonian dynamics and **Bayesian parameter estimation pipelines** for gravitational waves.

Employment History

Marie Curie fellowship under the grant No. 101208914. October 2025 - Ongoing
At the Niels Bohr Institute, Copenhagen.
In the Center of Gravity.

Postdoctoral fellowship in theoretical astrophysics. October 2023 - September 2025
At the Niels Bohr Institute, Copenhagen.
In the theoretical astrophysics group.

Ph.D position in theoretical astrophysics. 2019 - 2023
At the University of Zürich.
Under the supervision of Lucio Mayer (*lmayer@physik.uzh.ch*).

Teaching assistant in mathematics and physics. 2015 - 2019
At the Eidgenössische Technische Hochschule Zürich.

Private tutor for high-school students in mathematics. 2015 - 2018
For Zürcher Nachhilfe (*www.zuercher-nachhilfe.ch*).

Mountain Guide Assistant in Finale Ligure, Italy. 2013 - 2014
For Luigi Gagliardi (*gigio.gagliardi@gmail.com*).

Teaching, Supervision & Outreach

Private tutor for several Gymnasium students, at Zürcher Nachhilfe.

Teaching assistant at ETH in various mathematics and physics courses.

Supervision and support to Ph.D students at the Niels Bohr Institute, resulting in several shared publications.

Teaching assistant at UZH in theoretical Astrophysics and Cosmology, as part of my Ph.D duties.

Supervisor for the semester thesis of ETH student **Jeremy Layan**, on the topic of Post Newtonian expansions.

Supervisor for the semester thesis of ETH student **Marcus Haberland**, on the topic of exoplanets and gravitational waves. Accepted for Ph.D programme in Potsdam, Germany.

Actor and presenter at the "Three black holes walk into a bar" outreach event, in the Kosmos Klub, Zürich.

Interviewee in the hour long youtube video "Practically FREE Primordial Gravitational Waves Detector" by Fraser Cain, with over 35'000 views and overwhelmingly positive feedback.

Proposals, Workshops, Memberships

Member of the LISA Consortium.

Chapter Coordinator and Author for the LISA astrophysics working group white paper "Astrophysics with the Laser Interferometer Space Antenna".

Chapter Coordinator and Author for the TianQin astrophysics white paper.

Contributor to LISA's astrophysics "red book".

Co-Lead author of the accepted proposal "Future Missions to Uranus and Neptune: Prospects for Non-Planetary Science". ISSI, International Teams in Space and Earth Sciences.

Organiser of the workshop on "New Ideas on the origin of Black Hole Mergers". August 2024
At the Niels Bohr Institute.

Participant in the workshop on "scientific computing with Python". August 2022
At the University of Zürich, Zürich.

Participant in the workshop on "black hole dynamics". June 2022
At the Niels Bohr institute, Copenhagen.

Participant in the BINARY22 programme. May 2022
At the Kavli Institute for Theoretical Physics, Santa Barbara CA.

Participant in the workshop on "gravitational wave astronomy". August 2021
At the Niels Bohr institute, Copenhagen.

Selected and Recent Talks

Talk: Environmental effects on gravitational waves. December 2023
RESCEU workshop on black hole dynamics, Tokyo.

Talk: Imprints of accretion discs physics on gravitational waves. January 2023
Getting ready to descend the slippery slope of multimessenger black hole data, Sexten.

Talk: Direct collapse of exceptionally heavy black holes in the merger-driven scenario. October 2022
Young astronomers and galactic nuclei, San Sebastian.

Talk: Ice Giant Missions as Gravitational Wave Detectors. October 2021
LISA community Call (Virtual).

Invited talk: Progress in EMRI waveform modeling: Astro & GR September 2025
Università Insubria, Como.

Invited talk: Environmental effects in stellar mass GW sources March 2025
Sexten Center for Astrophysics

Invited talk: 2-D imaging with gravitational waves? September 2024
Università Insubria, Como.

Invited talk: Traces of accretion disc physics in gravitational waves. October 2022
Donostia International Physics Centre, San Sebastian.

Invited Talk: Traces of accretion disc physics in gravitational waves. May 2022
Kavli institute for theoretical physics, Santa Barbara CA.

Invited talk: The handbook of the gravitational wave astronomer. January 2022
GWNEXT conference, Beijing (Virtual).

Publication History up to October 2025

Summary

Total published works: 39
Excluding large collaboration papers: 36
First (or second) author publications: 14 (25)
Total citations: 1657
Excluding large collaboration papers: 632
h-index: 17
Excluding large collaboration papers: 15

Selected publications by topic

Gravitational waves and environmental effects

- Dirty Waveforms: multiband harmonic content of gas-embedded gravitational wave sources.*
Lorenz Zwick, Andrea Derdzinski, Mudit Garg, Pedro R. Capelo and Lucio Mayer.
Monthly Notices of the Royal Astronomical Society, Volume 511, Issue 4. April 2022
- The imprint of gas on gravitational waves from LISA intermediate-mass black hole binaries.*
Mudit Garg, Andrea Derdzinski, **Lorenz Zwick**, Pedro R. Capelo and Lucio Mayer.
Monthly Notices of the Royal Astronomical Society, Volume 517, Issue 1. September 2022
- Priorities in gravitational waveform modelling for future space-borne detectors: vacuum accuracy or environment?*
Lorenz Zwick, Pedro R. Capelo and Lucio Mayer.
Monthly Notices of the Royal Astronomical Society, Volume 521, Issue 3. May 2023
- Novel category of environmental effects in gravitational waves from binaries perturbed by periodic forces.*
Lorenz Zwick, Christopher Tiede, Alessandro A. Trani, Andrea Derdzinski, Zoltan Haiman, Daniel J. D'Orazio, Johan Samsing.
Physical Review D 110, 103005 November 2024
- Eccentric Features in the Gravitational-wave Phase of Dynamically Formed Black Hole Binaries.*
Kai Hendriks, **Lorenz Zwick** and Johan Samsing.
The Astrophysical Journal, Volume 985, Number 2 May 2025
- Dissecting environmental effects with eccentric gravitational wave sources.*
Lorenz Zwick, Kai Hendriks, David O'Neill, János Takátsy, Philip Kirkeberg, Christopher Tiede, Jakob Stegmann, Johan Samsing and Daniel J. D'Orazio.
Physical Review D 112, 063005 September 2025
- Environmental Effects in Stellar Mass Gravitational-wave Sources I: Expected Fraction of Signals with Significant De-phasing in the Dynamical and Active Galactic Nucleus Channels.*
Lorenz Zwick, János Takátsy, Pankaj Saini, Kai Hendriks, Johan Samsing, Christopher Tiede, Connar Rowan, and Alessandro A. Trani.
The Astrophysical Journal, Volume 991, Number 2 September 2025

Relativistic dynamics

- Improved Gravitational Radiation Timescales: significance for LISA and LIGO-Virgo sources.*
Lorenz Zwick, Pedro R. Capelo, Elisa Bortolas, Lucio Mayer and Pau Amaro-Seoane.
Monthly Notices of the Royal Astronomical Society, Volume 495, Issue 2. June 2020
- Relativistic binary-disc dynamics and the timing of OJ-287's flares.*
Lorenz Zwick and Lucio Mayer.
Monthly Notices of the Royal Astronomical Society, Volume 526, Issue 2 December 2023
- Close Encounters of Wide Binaries Induced by the Galactic Tide: Implications for Stellar Mergers and Gravitational-wave Sources.*
Jakob Stegmann, Alejandro Vigna-Gómez, Antti Rantala, Tom Wagg, **Lorenz Zwick**, Mathieu Renzo, Lieke A. C. van Son, Selma E. de Mink, and Simon D. M. White.
The Astrophysical Journal Letters, Volume 972, Number 2 September 2024
- Importance of relativistic pericenter precession in identifying the presence of a third body near eccentric binaries.*
Pankaj Saini, **Lorenz Zwick**, János Takátsy, Connar Rowan, Kai Hendriks, Gaia Faj, Daniel J. D'Orazio, Johan Samsing.
arXiv:2508.17348 August 2025

Formation and appearance of high redshift quasars

Direct collapse of exceptionally heavy black holes in the merger-driven scenario.

Lorenz Zwick, Lucio Mayer, Lionel Haemmerlè and Ralf S Klessen.

Monthly Notices of the Royal Astronomical Society, Volume 518, Issue 2.

January 2023

Direct Formation of Massive Black Holes via Dynamical Collapse in Metal-enriched Merging Galaxies at $z = 10$: Fully Cosmological Simulations.

Lucio Mayer, Pedro R. Capelo, **Lorenz Zwick** and Tiziana di Matteo.

The Astrophysical Journal, Volume 961, Number 1

January 2024

The evolution of accreting population III stars at 10^{-6} – $10^3 M_{\odot}/\text{yr}$.

Devesh Nandal, **Lorenz Zwick**, Daniel J. Whalen, Lucio Mayer, Sylvia Ekström and Georges Meynet

Astronomy & Astrophysics, 689, A351

February 2024

Little Red Dots as self-gravitating discs accreting on supermassive stars: Spectral appearance and formation pathway of the progenitors to direct collapse black holes.

Lorenz Zwick, Christopher Tiede and Lucio Mayer

arXiv:2507.22014

July 2025

Alternative gravitational wave detection methods

Imprints of massive black hole binaries on neighbouring deci-Hz gravitational-wave sources.

Jakob Stegmann*, **Lorenz Zwick***, Sander M. Vermeulen, Fabio Antonini, Lucio Mayer.

Nature Astronomy - 24010097B

July 2024

Bridging the micro-Hz gravitational wave gap via Doppler tracking with the Uranus Orbiter and Probe Mission: Massive black hole binaries, early universe signals and ultra-light dark matter.

Lorenz Zwick, Deniz Soyuer, Daniel J. D’Orazio, David O’Neill, Andrea Derdzinski, Prasenjit Saha, Diego Blas, Alexander C. Jenkins, Luke Zoltan Kelley.

Physical Review D 112, 083029

October 2025

Cosmology – Dark matter – Lensing

Prospects for a Local Detection of Dark Matter With Future Missions to Uranus and Neptune.

Lorenz Zwick, Deniz Soyuer and Jozef Bucko.

Astronomy and Astrophysics, Volume 664.

July 2022

Gravitational Wave Memory Imprints on the CMB from Populations of Massive Black Hole Mergers.

Lorenz Zwick, David O’Neill, Kai Hendriks, Philip Kirkeberg and Miravet-Tenés

Astronomy & Astrophysics, 694, A95

October 2024

The Proper Motion of Strongly Lensed Binary Neutron Star Mergers in LIGO/Virgo/Kagra Can Be Constrained by Measuring Doppler-induced Gravitational-wave Dephasing.

Lorenz Zwick and Johan Samsing

The Astrophysical Journal, Volume 988, Number 2

July 2025

Amplified capture rate of dark matter in compact binaries and constraints on bosonic dark matter from GW170817.

János Takátsy, **Lorenz Zwick**, David O’Neill and Johan Samsing

arXiv:2509.03272

September 2025